

Reza Golkar

AI Researcher | Machine Learning Engineer | M.Sc. in Artificial Intelligence

ABOUT

Focus

Artificial Intelligence, Machine Learning

Work Experience

+2 Years

Education

M.Sc. in Artificial Intelligence (Graduated)

M.Sc. in Artificial Intelligence (Degree Award Pending), graduated with an overall GPA of 19.65/20. Successfully completed and defended a master's thesis in Explainable Artificial Intelligence (XAI) for medical image analysis, achieving a thesis grade of 18.5/20.

Hands-on experience in Machine Learning, Deep Learning, Computer Vision, and Explainable AI, with a strong focus on developing interpretable AI systems for real-world healthcare applications.

Strong foundation in machine learning, including supervised and unsupervised learning, regression and classification models, optimization techniques, ensemble methods, and practical implementation using Scikit-learn and PyTorch.

Research experience in explainable deep learning models, including CNNs, EfficientNet, and Vision Transformers (ViTs), utilizing XAI techniques such as Grad-CAM and SHAP for model interpretation, fairness assessment, and bias analysis across different subgroups.

Proficient with Large Language Models (LLMs) and the Hugging Face ecosystem, including fine-tuning with PEFT/LoRA and deploying local AI solutions.

Experienced in building enterprise-grade Retrieval-Augmented Generation (RAG) systems and optimizing end-to-end NLP pipelines.

Skilled in designing and implementing AI agents and autonomous workflows, including orchestration of LLM-powered systems, tool integration, and multi-step reasoning pipelines. Hands-on experience with automation platforms such as n8n for building scalable, event-driven AI workflows and integrating APIs, databases, and external services.

Developed practical AI agent pipelines integrating LLMs with external tools (APIs, databases, automation frameworks), enabling autonomous task execution and decision-making.

2+ years of practical software development experience in mobile and web applications, with experience in Android development using Kotlin and modern application architectures.

WORK EXPERIENCE



Software Engineer (Mobile Applications)

Sep 2023 - Sep 2024 / 1 yr 1 mo

Megashid: Industry 4.0, IoT, WSN and Industrial Automation

- Developed and maintained Android applications using Kotlin and Flutter and Dart.
- Implemented custom UI designs and animations using Flutter framework.
- Collaborated with cross-functional teams to optimize app performance and user experience.



Software Engineer (Mobile Applications)

Mar 2022 - Aug 2023 / 1 yr 6 mos

Terapico: A startup offering online tree planting service

- Developed a native Android applications for an online tree planting service.
- Worked closely with product and backend teams to design scalable features and improve user engagement.
- Conducted code reviews and optimized app performance for diverse Android devices

EDUCATION

- M.Sc. in Artificial Intelligence Engineering**
Islamic Azad University, Khomeini Shahr
2024 – 2026
- B.Sc. in Information Technology Engineering**
Islamic Azad University, Najafabad
2012 – 2017

PREPRINTS

- Fair and Explainable Deep Learning for Skin Lesion Classification**

End-to-end pipeline using EfficientNet-B0, Grad-CAM, and fairness-aware evaluation for multi-class skin lesion classification across Fitzpatrick skin types.

SELECTED PROJECTS

XAI-Fair-SkinLesion-Classification

- Fine-tuned EfficientNet and ViT on 10,000+ dermatoscopic images, boosting accuracy by 8% and recall by 12%.
- Expanded dataset 4x via augmentation, improving F1-score for rare classes by 15%.
- Achieved 89% accuracy and real-time inference, with ViT outperforming ResNet50 by 7% in F1-score.

RAG Chatbot (Local & Secure)

- Developed a fully local and secure RAG chatbot for private company documents, eliminating external API dependency and reducing data leakage risk to zero.
- Implemented a modular RAG pipeline (ingestion, chunking, embedding, retrieval, generation), supporting 3+ document formats and multiple retrieval strategies.
- Achieved a 35% reduction in hallucinated responses by enforcing source-grounded answer generation.

Multilingual PDF Summarizer (Local LLM)

- Built a local LLM-based pipeline to summarize PDFs of up to 300 pages, reducing reading time by approximately 70%.
- Integrated NLLB for multilingual translation, achieving fluent Persian summaries with minimal semantic drift in qualitative evaluation.
- Implemented end-to-end NLP workflow (PDF parsing → chunking → summarization → translation) using PyTorch and Hugging Face.

SemanticSearch-Chatbot – NLP-based QA

- Built a semantic search chatbot using SentenceTransformers (all-MiniLM-L6-v2) and FAISS, enabling sub-second retrieval over 10,000+ text chunks.
- Improved answer relevance by ~30% compared to keyword-based search using cosine similarity over dense embeddings.
- Developed an interactive Gradio interface, enabling rapid experimentation and reducing evaluation time by 40%.

Mytrees: Online tree planting service

- Developed the mobile client using Kotlin and Jetpack Compose, integrating REST APIs, Firebase services, authentication, and real-time updates for user and tree data.
- Collaborated with backend and product teams to design scalable features, apply MVVM architecture, and improve app reliability and performance in production environments.

LLm-interview-summarization-analysis

- Analyzed summarization behavior of LLMs on 50+ real-world interview transcripts, comparing zero-shot and few-shot prompting strategies.
- Observed up to 20% improvement in coherence and coverage using few-shot prompts compared to zero-shot baselines.
- Compared FLAN-T5 and BART models, highlighting trade-offs between factual consistency and abstraction quality.

LANGUAGES

English

Professional proficiency

German

B2 Certificated



Workout



Jogging



Dining out

SKILLS

AI Agents & LLM Systems

AI Agent Design, Autonomous Workflows, LLM Orchestration, Tool Integration (APIs, Databases, External Services), Multi-step Reasoning Pipelines, Retrieval-Augmented Generation (RAG), Prompt Engineering, Context Management, Agent-based Automation

Automation Platforms: n8n (workflow automation, event-driven pipelines, API integrations)

Machine Learning & Statistics

Gradient Descent, Logistic Regression, k-NN, k-Means, SVM, Decision Trees, Random Forest, Ensemble Learning, PCA, Evaluation Metrics (Accuracy, Precision, Recall, F1, AUC-ROC)

Deep Learning & AI

Deep Neural Networks, CNNs, Autoencoders, Transformers, Attention Mechanism, Vision Transformers (ViT), Regularization (L1/L2,Dropout),Optimization(SGD,Adam)

Frameworks: Scikit-learn, PyTorch, TensorFlow, TensorFlow Lite

Data & MLOps Mindset

Data Science (EDA, Data Visualization, Statistical Analysis), REST APIs, CI/CD Pipelines, Unit Testing, TDD, Agile/Scrum

Programming & Tools

Python, SQL, Git, Linux, VS Code, Kotlin

Software Engineering (Mobile)

Mobile Applications, Kotlin, Android Studio, Firebase, MVVM Architecture